

# Adoption, Usage, and Perceptions of LLM-Generated Feedback in Computer Science Education: A Cross-Cultural Study of Students and Academics

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**Abstract**—This full paper explores the use of Large Language Models (LLMs) for generating assessment feedback in Computer Science (CS) education, comparing adoption patterns and perceptions of students and academics in the UK and Saudi Arabia (KSA). The study investigates (1) differences in LLM use and adoption between students and academics in both countries, and (2) participants’ perceptions of LLM-generated feedback, focusing on benefits, challenges, and areas for improvement. A mixed-methods approach was employed, involving 251 participants. The collected data were analyzed through descriptive statistics and thematic analysis. Findings indicate that students in KSA demonstrate higher usage rates of LLMs, with ChatGPT identified as the most frequently used tool. Although KSA academics show limited use of LLMs, UK academics did not use them for feedback generation. Thematic analysis revealed three key themes: (a) *“It saves effort”*, (b) *“It is not as smart and genuine as a human”*, and (c) *“Making it more like humans and less like a machine”*. The findings emphasize the need for institutional policies to guide the responsible integration of LLM-generated feedback, ensuring alignment with teaching goals and improving feedback quality.

**Index Terms**—Computing education, Mixed methods research, Feedback, Higher education, perceptions, Large Language Model, Generative AI.

## I. INTRODUCTION

Large Language Models (LLMs) are AI systems designed to interpret and generate human-like language using vast amounts of text data [1], [2]. Based on transformer architectures [3], these models are able to capture complex language structures and perform a variety of tasks. Techniques such as in-context learning [2] and reinforcement learning from human feedback (RLHF) [4] improve the quality, clarity, and relevance of their outputs. ChatGPT, in particular, has emerged as a widely recognized example, demonstrating strong abilities in natural language generation and flexibility in handling a wide range of

tasks without task-specific training [5]. Although LLMs have reshaped natural language processing, their adaptable features are now gaining attention in the field of education. A recent survey [6] grouped their educational applications into three key areas: helping students (e.g., answering questions, correcting grammar), supporting instruction (e.g., grading, content generation) and enabling personalized learning (e.g., knowledge tracing and adaptive pathways). These uses highlight the growing potential of LLMs in enhancing teaching and learning in higher education, with increasing interest in their use for formative feedback as an emerging area of research.

Feedback is widely recognized as a fundamental element of effective teaching and learning in higher education [7]. It plays a vital role in helping students assess their performance, identify areas of growth, and make significant progress in their learning journey [8]. This is particularly important in computer science (CS) education, where effective feedback plays a central role in helping students develop coding skills and solve problems [9], [10]. However, providing timely and personalized feedback remains a significant challenge, particularly due to logistical constraints such as large class sizes [11], [12]. To address this gap, recent studies have explored the use of generative AI tools, particularly LLMs, to automate and improve feedback provision in computing education [13]–[22].

Although these efforts have made substantial contributions by investigating LLM-generated feedback in real classroom settings and evaluating its accuracy and usefulness, most of them focus mainly on the technical aspects and student experience during the actual use of the tools. Moreover, there is still a limited understanding of how institutional and cultural factors may shape key perceptions of LLM-generated feedback in computer science education, particularly regarding its usage,

adoption, perceived usefulness, and satisfaction.

This study addresses this gap by investigating the adoption, use, and perceptions of LLM-generated feedback in computer science education, employing a cross-cultural design that includes participants from two distinct higher education systems. The study aims to contribute to a deeper understanding of how LLM-generated feedback is integrated and experienced in practice, shedding light on both its perceived benefits and the challenges that may hinder its effective use.

This paper is structured as follows: Section II reviews the literature, Section IV describes the methodology, Section V presents the results, Section VI discusses the key findings, and Section VII concludes the paper.

## II. LITERATURE REVIEW

The growing use of generative AI tools in higher education has prompted researchers to investigate how students and educators perceive, adopt, and utilise these technologies, particularly in teaching, assessment, and feedback. The literature reflects a variety of perspectives shaped by institutional, cultural, and disciplinary factors.

In Saudi Arabia, several studies have demonstrated that students actively utilise tools like ChatGPT in their academic work. An extensive national study [23] reported that most students use generative AI to clarify concepts, summarize texts, translate content, and generate ideas. They found it helpful due to its speed and accessibility, but concerns were raised about subscription costs, unreliable output, and reduced learning independence. Similarly, a study of healthcare students [24] found high awareness of ChatGPT and generally positive attitudes. Many students believed that it improved their productivity and helped them learn more effectively. Among students of English as a Foreign Language (EFL), [25] highlighted that ChatGPT was considered moderately effective in improving language understanding and communication, especially due to its ease of use. Students were more willing to continue using it when they found it convenient and straightforward. [26] explored its use in writing courses, noting perceived benefits such as increased creativity, better learning experiences, helpful academic support, and increased productivity. However, the students also pointed out risks such as a lack of human tone, potential plagiarism, and anxiety about relying too much on AI.

In the UK, [27] found that computer science students frequently used ChatGPT, often on a weekly basis, to support their learning. They used it to access information, understand problems, and work more efficiently. Although students believed it was fair to use AI tools in academia, many were concerned about plagiarism and reported moderate trust in the accuracy of the tool. The study emphasized the need for students to have clear guidelines on responsible use. [28] found that postgraduate researchers used generative AI as a time-saving tool, a writing assistant, or a collaborative partner. Most users accepted its practical value, but there was uncertainty about its limitations and ethical boundaries. In another study [29], interviews with students and instructors

in computing education revealed shared views on the benefits of generative AI to improve learning and reduce workload. However, both groups raised concerns about overreliance and the trustworthiness of AI tools. The findings highlighted the need for curriculum adjustments that support the effective integration of AI tools while addressing potential risks such as plagiarism and enhancing equitable access to educational support.

Internationally, several studies offered comparative or multi-country insights. [30] compared students and educators in Australia, Cyprus, and the United States to evaluate the quality of online assessment. Both groups viewed essay and coding tasks as the most impacted by generative AI. Educators were more likely to suggest adapting assessments to encourage higher-order thinking, while students had mixed reactions, including concerns about reduced creativity. In another study, educators highlighted the value of using AI to support learning processes, but emphasized the importance of ethics, human interaction, and focusing on how students learn [31]. In the Middle East, [32] investigated how university instructors approached generative AI for assessment. Instructors reported positive views on generative AI because it helps reduce marking workload and increase student engagement. Many also described redesigning assignments to more effectively integrate AI tools. However, concerns were raised about the potential harm to student writing abilities and critical thinking. In the United States, [33] surveyed 328 college students and showed that ChatGPT's efficiency and compatibility positively influenced its ease of use, usefulness, and students' intention to keep using it, based on the Technology Acceptance Model (TAM).

Some studies also explored LLMs as feedback providers. [20] showed that students generally viewed ChatGPT-generated feedback on programming tasks as helpful and aligned with formative feedback principles. The students preferred feedback that included their code as it was more precise. The study also addressed practical concerns, including and more relevant. However, preferences varied with respect to the tone of feedback. Similarly, [16] presented a large-scale study in a global CS1 course where real-time style feedback generated by LLMs improved student engagement and prompted significant improvements in code quality. Students who received instant feedback were significantly more likely to revise their code, and the study also addressed practical concerns such as consistency, fairness, and demographic bias. In a more focused context, [19] examined how GPT-4 generated hints for compiler errors in an introductory programming course. Students reported reduced frustration and increased focus when receiving GPT-generated hints, and feedback was generally perceived as useful. Although performance gains were mixed depending on the error type, students still outperformed the control group in later stages without hints, suggesting a learning effect from earlier feedback.

Recent studies in Computing Education Research (CER) have highlighted several benefits of using LLM-generated feedback in higher education. These include the ability to

deliver personalized code correction [18], tiered or multilevel feedback to support students at different stages of learning [14], and real-time feedback on coding style [16]. Other studies have shown the usefulness of LLMs in generating automated next-step hints [15], responding to student help requests [13], offering consistent and reliable feedback [17], and improving students' ability to resolve syntax-related errors [19]. Despite these advantages, concerns persist regarding the reliability of AI-generated feedback. Several studies [13], [15], [18] have reported issues with inaccurate suggestions or hallucinated errors. Furthermore, ensuring that the feedback is adapted to the level and context of each student, often referred to as context-sensitive feedback, remains a challenge [16].

In summary, research on LLM-generated feedback in CS education remains limited, particularly across diverse cultural and academic contexts. This study addresses these gaps by exploring how such feedback is adopted, used, and perceived in two distinct higher education settings.

### III. RESEARCH QUESTIONS

In light of the focus of the study, the following research questions were formulated.

**RQ 1:** How do students and instructors in computer science education in the UK and KSA use, perceive, and intend to adopt LLM-generated feedback?

**RQ 2:** What are the perceived benefits, challenges, and suggestions to improve the use of LLM-generated assessment feedback in CS education in the UK and KSA?

### IV. METHODOLOGY

This study used a mixed-methods survey design [34] to examine the use of LLMs to generate formative feedback on computer science assessment tasks. Closed-ended items provided measurable data, while open-ended questions allowed participants to elaborate on their views, offering richer insights into perceived benefits, challenges, and future intentions associated with LLM-based feedback practices [35].

#### A. Participants

A total of 251 participants were involved in the study, comprising 174 students and 77 instructors from public universities in the UK and KSA. All participants came from disciplines related to computing, including computer science, software engineering, and information technology, at both the undergraduate and postgraduate levels. Data were collected during the first semester of the academic year 2024-2025, a period marked by the increasing use of generative AI tools in education, including GPT-4o, Gemini and QuillBot.

#### B. Survey Instrument

Two online questionnaires were developed: one for students and one for instructors. They were adapted from a semi-structured interview guide used in a previous study on generative AI tools in higher education [29], specifically focusing on

the use of LLMs to generate feedback on computing assessment tasks. The surveys included multiple-choice and Likert scale items covering LLM usage frequency, satisfaction, and perceived usefulness, along with four open-ended questions asking about experiences of using LLMs to generate feedback. To accommodate participants from both regions, the surveys were available in English and Arabic. Before deployment, a pilot study was conducted with seven participants (students and educators) to ensure clarity and usability.

#### C. Procedure

Participants were recruited through institutional mailing lists during the Fall 2024 semester and invited to complete an online survey. Participants were informed that participation was entirely voluntary and anonymous, with the option to withdraw at any time without any consequences. Ethical approval was granted by the Ethics Committee at the University of Glasgow and the Deanship of Scientific Research at Imam Abdulrahman Bin Faisal University in Saudi Arabia.

#### D. Data Analysis

Quantitative data were processed using IBM SPSS Statistics (version 29). Given the exploratory nature of the study, descriptive statistics such as frequencies and percentages were applied to summarize participant responses and identify key patterns. Qualitative data from open-ended questions were analyzed thematically following six-phase approach [36]. This process involved familiarizing with the data, generating initial codes, identifying and defining themes, reviewing and naming themes, and producing a final report. Coding was performed manually by the primary researcher, with independent review by a second researcher to enhance reliability. Disagreements were resolved through discussion.

### V. RESULTS

#### A. Demographic

The study included 174 students and 77 Academics from KSA and the UK (see Tables II and I). In KSA, most of the students were female (78.5%), while in the UK, most were male (69.2%). A small number of UK students (7.7%) preferred not to say their gender. Most students were in advanced years, with juniors and seniors forming the largest groups (40.7% and 23.0% in KSA; 20.5% and 25.6% in the UK). Graduate students made up 8.1% in KSA and 15.4% in the UK.

Among instructors, women were the majority in KSA (78.1%), while men were more common in the UK (61.5%). Most of the instructors in KSA were lecturers (43.8%) or assistant professors (34.4%). In the UK, 38.5% were lecturers and 38.5% were graduate teaching assistants (GTA).

#### B. Quantitative Results (RQ1)

The findings reveal distinct patterns in the use, adoption, and intention to use LLMs for assessment feedback across students and academics in both the KSA and the UK, with notable

TABLE I  
PARTICIPANT CHARACTERISTICS – ACADEMICS

| Demographics            | KSA (n=64) | UK (n=13) |
|-------------------------|------------|-----------|
| <b>Gender</b>           |            |           |
| Male                    | 14 (21.9%) | 8 (61.5%) |
| Female                  | 50 (78.1%) | 5 (38.5%) |
| Prefer not to say       | 0 (0%)     | 0 (0%)    |
| <b>Teaching Role</b>    |            |           |
| TA/GTA                  | 7 (10.9%)  | 5 (38.5%) |
| Lecturer/Instructor     | 28 (43.8%) | 5 (38.5%) |
| Sr. Lecturer/Asst. Prof | 22 (34.4%) | 2 (15.4%) |
| Assoc. Prof/Professor   | 7 (10.9%)  | 1 (7.7%)  |

TABLE II  
PARTICIPANT CHARACTERISTICS – STUDENTS

| Demographics         | KSA (n=135) | UK (n=39)  |
|----------------------|-------------|------------|
| <b>Gender</b>        |             |            |
| Male                 | 29 (21.5%)  | 27 (69.2%) |
| Female               | 106 (78.5%) | 9 (23.1%)  |
| Prefer not to say    | 0 (0%)      | 3 (7.7%)   |
| <b>Year of Study</b> |             |            |
| Freshman             | 19 (14.1%)  | 3 (7.7%)   |
| Sophomore            | 19 (14.1%)  | 12 (30.8%) |
| Junior               | 55 (40.7%)  | 8 (20.5%)  |
| Senior               | 31 (23.0%)  | 10 (25.6%) |
| Graduate             | 11 (8.1%)   | 6 (15.4%)  |

variations between participant group (students and academics) and regional contexts (UK and KSA).

#### Usage Frequency

As illustrated in Figure 1, students in both countries reported more frequent use of LLMs than instructors, though usage patterns differed. In KSA, 94.1% of students had used LLMs, and nearly half used them frequently (daily or several times a week). Among instructors, half reported frequent use, while a quarter had never used them. In the UK, student use was also high (89.7%), with just over half using LLMs frequently. However, daily use was less common, and many students reported using it only occasionally. UK instructors showed the lowest overall engagement: while three-quarters had used LLMs, only a small fraction (16.7%) used them daily, and more than half used them rarely or not at all.

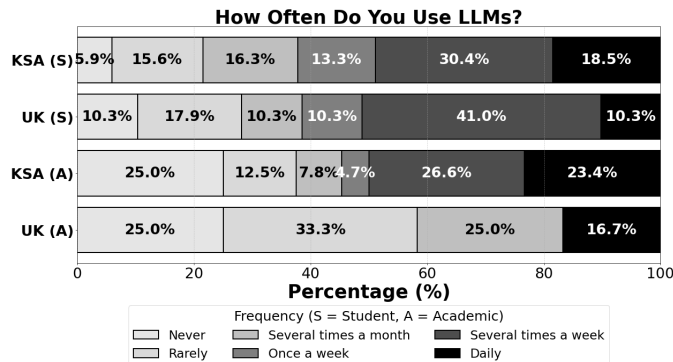


Fig. 1. Frequency of LLM usage by participant group and country

#### Tool Preference

As shown in Figure 2, ChatGPT was the most commonly used LLM among all participants. In KSA, 52.6% of students and 37.4% of instructors reported using it. In the UK, student usage was similar at 49.2%, while instructor usage was slightly higher at 38.5%. GitHub Copilot was slightly more used by UK students, while Microsoft Copilot was used the most by KSA students. Instructor use of these other tools was low and predominantly reported in KSA. Most instructors reported not using any LLM tool, especially in the UK, where 53.9% had never used one, compared to 26.4% in KSA. Among students, non-use of LLMs was low in both countries (KSA: 5.7%, UK: 10.2%).

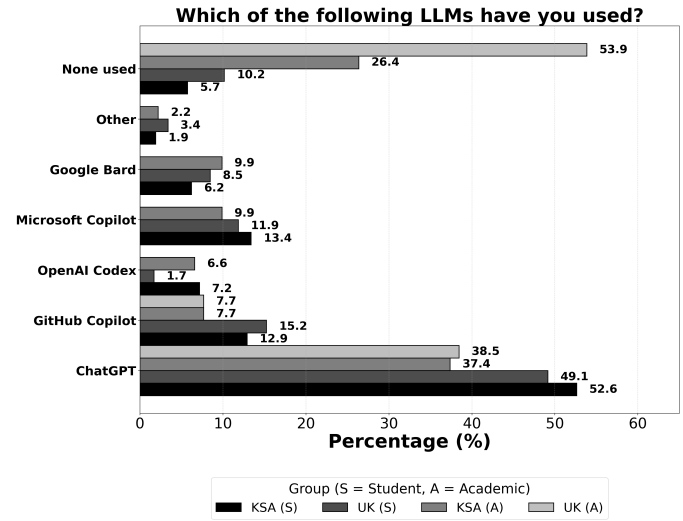


Fig. 2. LLM tool preference by participant group and country

#### Adoption of LLMs for Feedback Generation

As shown in Figure 3, students in both countries were more likely than instructors to use LLMs to generate feedback on assignments. In KSA, 66.7% of the students reported doing so, compared to 20.3% of the instructors. In the UK, more than half of the students had used LLMs for feedback, while none of the instructors reported any use. In KSA, 79.7% of the instructors had not used LLMs for feedback. Among these non-users, 50.9% were lecturers (26 out of 51), and 39.2% were senior lecturers or assistant professors (20 out of 51). Teaching assistants represented a smaller portion of non-users. On the student side, the adoption of LLM in KSA was highest among third-year students, with 72.7% (40 out of 55) reporting usage. The first and second year students had a combined adoption rate of 52.6%. These patterns highlight differences in adoption rates between roles and academic levels, with higher usage among students and lower usage among instructors, particularly those in mid- to senior-level positions.

#### Satisfaction levels with LLM-generated feedback

As shown in Figure 4, satisfaction levels varied between groups and countries. In KSA, most of the students reported positive experiences with LLM-generated feedback. A total of 62.1% said that they were very or extremely satisfied

Have you used LLMs to get feedback on your /your students' assignments?

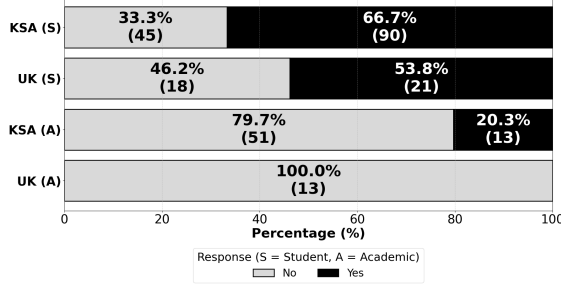


Fig. 3. Adoption of LLMs for feedback generation by participant group and country

and 34.5% were moderately satisfied. Only 3.4% reported low satisfaction. Overall satisfaction among UK students was lower, with 52.7% indicating they were very or extremely satisfied, and 36.8% reporting moderate satisfaction. Their dissatisfaction was higher compared to KSA, with 10.6% reporting low satisfaction.

Among KSA instructors, 75.0% were very or extremely satisfied (41.7% and 33.3%, respectively), and the remaining 25.0% were moderately satisfied. None reported dissatisfaction. In summary, satisfaction with LLM-generated feedback was higher in KSA than in the UK, especially among students and instructors who had used the tools. No satisfaction data was collected from UK instructors, as none reported using LLMs for feedback generation.

How satisfied are you with the feedback generated by LLMs?

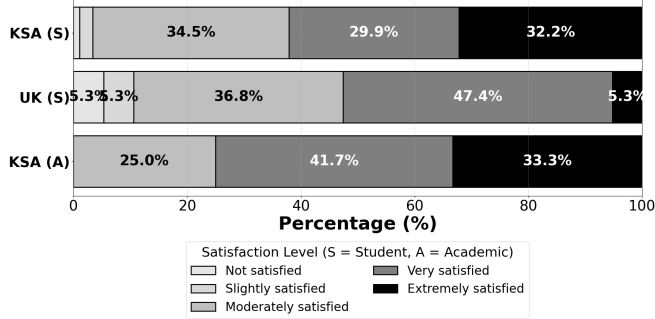


Fig. 4. Satisfaction with LLM-generated feedback by participant group and country

**Helpfulness of LLM-generated feedback** Participants rated the helpfulness of LLM-generated feedback. As shown in Figure 5, most students in KSA (73.6%) found the feedback very or extremely helpful. Only 1.1% considered it not helpful, while 10.3% rated it as slightly helpful and 14.9% as moderately helpful, indicating a generally high level of perceived usefulness. In contrast, responses from UK students were more varied. While 52.6% found the feedback very or extremely helpful, 36.8% rated it as moderately helpful, and 10.5% as slightly helpful. Notably, none considered it unhelpful. KSA instructors also reported favorable perceptions, with 75% rating the feedback as very or extremely helpful and 25% as moderately helpful. None rated it as unhelpful.

How helpful do you find the feedback generated by LLMs?

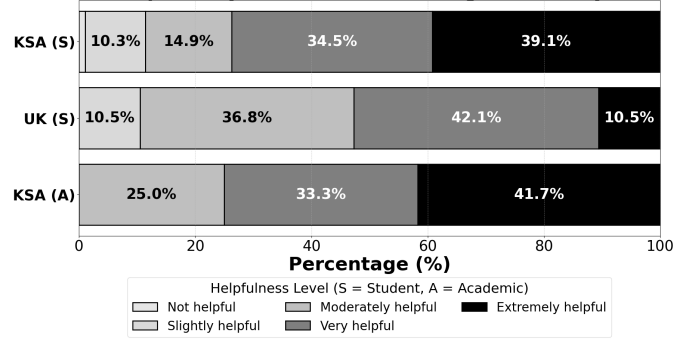


Fig. 5. Helpfulness of LLM-generated feedback by participant group and country

**Intentions to Use LLMs for Feedback Generation** As shown in Figure 6, intentions to use LLMs for feedback generation varied across groups and countries. In KSA, 64.4% of students expressed interest in future use, compared to only 5.6% in the UK. Among instructors, the pattern was more balanced, with 34.0% in KSA and 33.3% in the UK indicating interest. These results suggest a stronger openness to adopting LLMs among Saudi students in particular, whereas UK students appeared more resistant, with 94.4% reporting no intention to use such tools.

Do you plan to use LLMs to generate feedback on your /your students' assignments?

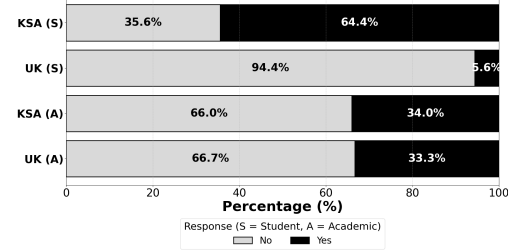


Fig. 6. Intentions to use LLMs for feedback generation by participant group and country

### C. Qualitative Results (RQ2)

A thematic analysis of participants' qualitative responses revealed three overarching themes regarding LLM-generated feedback in computer science education across the UK and KSA: (a) "It saves effort", (b) "It is not as smart and genuine as a human", and (c) "Making it more like humans and less like a machine". These themes offer insight into how students and educators in both regions assess the role of LLMs in the feedback process, the concerns they have about its limitations, and the ways they envision enhancing its value in teaching and learning within computer science education.

**1) "It Saves Effort":** This theme reflects the perceived practical value of LLM-generated feedback, particularly its time-saving and academic support functions. It was the most prominent theme across both KSA and UK participants, who emphasized its usefulness, clarity, and efficiency. Several sub-themes highlighted specific aspects of this value.

a) *Efficiency and Speed*: One of the most frequently mentioned benefits was the ability of LLMs to provide instant feedback and save time. This feature was appreciated by students dealing with tight deadlines, as well as educators who manage heavy workloads. These responses illustrate a shared perception between regions and roles about the time-saving and efficiency-enhancing potential of LLM-generated feedback.

For example, a KSA teacher noted, *"Save time, can describe what I want in short and clear words quickly, save my effort for doing hard job."* Another KSA teacher emphasized the tone and delivery of the feedback as a key reason for using LLMs: *"It formulates and delivers feedback gently and focuses on the positive aspects, as some people do not accept criticism."* Similarly, a UK teacher shared, *"it saves my time and allows me to give detailed feedback quickly that helps students understand and fix their mistakes, improving their skills. It also frees up my time to focus on other important tasks in the course."*

The students also emphasized this benefit. A UK student commented that *"LLMs can act as instant tutors and often explain the whys of particular reasonings and conclusions."* and a student at KSA highlighted *"Getting sources faster by shortening the time spent searching online."*

b) *Detail and Clarity*: The participants reflect a common appreciation for the way LLM can enhance the depth and clarity of educational feedback. This was especially appreciated by students who found that the feedback from lecturers or teaching assistants was often too brief. As one UK student commented, *"They are more detailed than some of my lecturers or teaching assistants."* Educators also recognized this benefit in enhancing the quality of their feedback. A UK teacher stated *"I think I will use it to make my feedback more clear and without grammar mistakes."* A KSA teacher agreed with this view, saying, *"To analyze answers more thoroughly and get suggestions that could help develop the student."* Students in KSA similarly valued the level of detail and clarity provided by LLMs. One student noted, *"Due to the fact that it provides as many details as the person needs, it also provides images and examples to demonstrate, and very clear."*

c) *Error Spotting*: Student participants highlighted the role of LLMs in detecting errors, particularly in programming assignments. This feature was viewed as useful for self-assessment and revision. A KSA student explained, *"It helps me identify mistakes very quickly, for example, when I write code and there is an error, it helps me locate the mistake."* Similarly, a UK student noted, *"Can sometimes spot mistakes."*

d) *Improved Understanding*: LLMs were also perceived as effective tools to improve student understanding. Participants noted that these tools not only highlighted mistakes but also explained underlying concepts, helping learners identify areas for improvement. A UK student remarked, *"I learn much better when it explains it to me and tells me where I went wrong and how to improve."* A KSA student shared a similar view: *"To get an initial idea of the kind of feedback I might*

*receive from the course instructor, and to see if there are things I can improve before submitting."* Additionally, a KSA teacher emphasized the learning potential of LLMs, stating, *"I use it myself to look for suggestions to improve my work. I found it very helpful, so I plan to use it for the benefit of students."*

e) *Support for Coding and Projects*: Finally, LLMs were frequently described not only as a feedback tool but also as a supportive tool to shape and refine academic work. The students used them to brainstorm ideas and organize their thoughts, especially when struggling to express complex concepts. As a UK student shared, *"I use it to help me brainstorm ideas like discussing with another person instead of seeing it as a tool..."* A KSA student highlighted a similar benefit: *"To improve the quality of my work and identify missing parts of the project."* Educators also acknowledged this role. A KSA teacher stated, *"Helps me make necessary improvements, whether in student projects, research, or exams."*

2) ***"It is not as smart and genuine as a human"***: This theme reflects the concerns and perceived limitations of using LLMs for feedback. Although participants acknowledged the potential value of the system, they also raised concerns related to precision, depth, ethical use, and the human-like quality of the responses. Both students and educators expressed caution about relying too heavily on such tools.

a) *Accuracy and Clarity*: A shared concern in both regions was the reliability and clarity of the LLM-generated feedback, particularly in academic settings where precision is critical. This concern was especially emphasized by UK students who reported that responses were sometimes vague, misleading, or overly confident. One UK student noted, *"It doesn't always know the answer and can take you in the wrong direction."* Another added, *"The LLM could mark something as incorrect, but would not be able to report to the student where the flaw in their understanding is."* A KSA student also highlighted clarity issues, stating that *"uses complex and unusual words, and its style is unique."* Teachers expressed similar concerns. A KSA teacher remarked that *"It does not always provide correct answers."* A UK teacher commented on the use of LLM in research contexts, saying, *"Work with them for research – they are unreliable."*

b) *Prompting Difficulty*: Several participants reported that generating useful and accurate feedback from LLMs required highly specific and detailed instructions, which created a barrier to effective use. A student in the UK explained: *"I need to carefully craft the prompts because if you did not provide enough detail, often the answers are plain incorrect."* Another added, *"sometimes it takes too long to give it all the context."* Participants in KSA reported similar difficulties. A teacher shared, *"Challenge in trying to explain to the AI and help it understand my intent."* A student noted, *"It not comprehending the content I shared with it to make it give me complete and correct feedback."*

c) *Cost*: The financial barrier to accessing premium versions of LLM was another concern, especially for students. A UK participant stated, *"Paying for it."* Similarly, a KSA student similarly noted, *"Costly."*

d) *Institutional and Ethical Concerns*: The participants' responses about using LLMs for student work reflect a variety of ethical and institutional concerns, including academic integrity, data privacy, and discomfort with the broader AI industry. As one UK teacher explained, *"For ethical reasons I would not enter a student's submitted work in an LLM tool unless I know for sure that work won't be used by that LLM for training."* Another added, *"Auto-generating feedback seems like a betrayal of my duty as an educator."* Similarly, a KSA teacher commented, *"The feedback is about humans not machines, so I prefer to give feedback myself rather than get help from LLM."* Some students also expressed skepticism regarding the legitimacy or ethics of AI feedback. A UK student remarked, *"Morally opposed to LLM."* Another explained: *"I find the AI (LLM) industry icky and I don't want to use their tools, companies openly steal and take traffic on many websites to download their data without consent, and I can't imagine myself ever being comfortable supporting."* Similarly, a student of KSA stated *"I wish to use it, but my university considers it cheating."*

3) *Making It More Like Humans and Less Like a Machine*: A significant number of participants proposed improving LLMs to be more human-like, particularly in how they interpret, evaluate, and deliver feedback. These suggestions were often linked to dissatisfaction with responses that felt robotic or generic. A KSA student said, *"Feedback should be based on credible academic sources."* Another emphasized, *"Its responses should be reviewed by experts."* Similar concerns were raised by UK students, who called for greater reliability and logical accuracy: *"More factually correct feedback."* In addition, many participants believed that LLMs could be more effective if tailored to individual academic contexts, disciplines, or student needs. This included suggestions for personalized feedback, course-specific training, and better alignment with academic expectations. A UK student suggested, *"I think that LLMs should be trained in the specifics of a course."* Another noted, *"Have access to specialized LLMs that are trained for more specific purposes."* Similar requests were also raised in the KSA context. A student commented on *@Add customization options,* while a teacher proposed *"Adding more tools and more precise software to the AI."* The participants also shared practical recommendations for improving the overall user experience. For example, a KSA student stated *"Based on my experience LLM, tools, never give other tool name the machine just write theoretical book style answer without providing any name."*

## VI. DISCUSSION

This study explored the use and perceptions of LLM-generated feedback in computer science education in two different academic and cultural contexts: the UK and KSA. The following discussion interprets the key findings and considers how cultural and institutional factors may have influenced the results.

The findings show that students in both regions reported more active use of LLMs than instructors. This may be due

to the regular use of AI-based generative tools by students for coding and problem solving, as also highlighted in [29]. This may also reflect the students' perceptions of these tools as effective and easy to use. As noted in [33], the acceptance of LLMs such as ChatGPT is strongly influenced by perceived efficiency and usability, consistent with the Technology Acceptance Model (TAM). In addition, students in technical fields often demonstrate a greater openness to digital tools [37], which may explain their willingness to engage with LLM as a reliable source of academic support. In contrast, instructors reported lower overall usage, although those in KSA showed greater engagement than their UK counterparts. This difference may be partially shaped by national initiatives such as Vision 2030, which actively promote educational innovation and the adoption of emerging technologies in Saudi Arabia [38].

ChatGPT was identified as the most frequently used LLM, largely due to its effectiveness in supporting coding tasks. Its capabilities, including code generation, natural language interaction with databases, and integration with programming tools such as Python, make it a practical and accessible resource for both students and instructors [5].

Students in both regions reported actively using LLMs to obtain feedback on their assignments. The open responses described that the students used these tools to check their code, identify errors, and improve their work prior to submission. This behavior may reflect their perception of LLMs as practical and accessible tools that support independent learning, particularly in programming contexts [29]. Previous research also highlights the ability of LLMs to provide immediate and targeted feedback on syntax and logic issues, making them especially valuable to CS students during task completion [16], [19], [20]. By contrast, instructors demonstrated a more cautious stance toward using LLMs for feedback. In the UK, no one reported using such tools, reflecting ongoing concerns about academic integrity and plagiarism [31]. In [29], instructors further expressed concerns that students might not fully grasp key concepts or skills if they relied too heavily on LLMs, and they also highlighted the lack of clear institutional policies as a major source of uncertainty regarding how to integrate these tools into their teaching. However, in Saudi Arabia, limited adoption among instructors may be related to growing national support. The Ministry of Education has issued official guidelines encouraging the regulated use of generative AI in education, including its application for feedback provision [39]. Another key finding relates to how participants evaluated the usefulness and helpfulness of LLM-generated feedback. UK instructors were excluded from this analysis, as none reported using LLMs for CS feedback. In KSA, overall satisfaction and perceived helpfulness were high, possibly influenced by greater familiarity with LLMs, openness to educational technology, and national support, such as the role of the Saudi Data and Artificial Intelligence Authority (SDAIA) [40]. These efforts, along with the Ministry of Education guidelines that promote responsible AI use, can enhance confidence and engagement [39]. In contrast, lower

satisfaction with LLM-generated feedback among UK students may be explained by limited experience with LLMs and uncertainty about how to use them effectively. Previous studies show that students with less familiarity tend to be more passive and less satisfied [28], and that satisfaction is influenced by ease of use and perceived efficiency [33]. Ethical concerns may also contribute to more cautious adoption in the UK [27].

The findings revealed a generally low intention to use LLM-based tools for academic assignments, particularly among students in the UK. This hesitation offers further insight into the institutional and cultural dynamics that may shape the attitudes of participants towards generative AI. Previous studies suggest that students in the UK often report uncertainty about what constitutes acceptable or ethical use of generative AI tools, such as ChatGPT [27], [28]. This uncertainty may be reinforced by the lack of clear institutional policies, training, or academic guidance, leaving students hesitant to apply such tools in formal coursework [28], [31]. As a result, even students who recognize the usefulness of LLM-generated feedback may avoid using it in graded tasks to prevent potential breaches of academic integrity or failure to comply with unclear institutional guidelines. This cautious approach contrasts with the greater willingness observed in contexts like KSA, where institutional openness and policy support have created a more encouraging environment for responsible and guided use of generative AI [38]–[40].

Thematic analysis of open-ended participants' responses revealed nuanced perspectives on LLM-generated feedback in CS education. The most prominent theme, *It saves effort*, reflects a widely shared appreciation of the speed, clarity, and usefulness of such tools, particularly in programming contexts. These findings align with previous research that highlights the ability of LLMs to improve efficiency, support creativity, and enhance student engagement through timely and detailed feedback [16], [19], [20], [23], [26], [33]. At the same time, participants raised concerns under the theme *It is not as smart and genuine as a human*, including issues related to reliability, vagueness, and ethical ambiguity. These concerns echo findings from earlier studies on the risks of academic misuse, the absence of human judgment, and the need for clearer institutional policies to guide responsible adoption [24], [24], [26]–[28], [31]. Finally, the theme *Making it more like humans and less like a machine* underscores the desire of participants for more personalized, context-sensitive and academically aligned feedback. This is consistent with the recommendations in the literature to improve accuracy, personalization, and consistency in LLM-driven feedback systems [16], [19], [20], [31].

## VII. PEDAGOGICAL IMPLICATIONS AND CONCLUSION

This study investigated the usage, adoption, and perceptions of LLM-generated feedback among CS students and academics in the UK and KSA. The findings highlight the importance of institutional support, including clear ethical guidelines and training to ensure the effective and responsible use of these tools [28], [31]. Training should include practical

classroom examples to help users critically evaluate and reflect on LLM-generated feedback.

LLMs should be positioned as supportive learning tools rather than evaluative ones. To achieve this, educators must carefully craft prompts that guide the LLM to provide scaffolded feedback (e.g., step-by-step explanations) [41] and include gamified or interactive elements to improve engagement and reduce anxiety [14]. Prompts should also foster students' critical thinking and self-reflection, encouraging them to revise their work and engage more deeply with feedback [23]. However, if poorly constructed, such prompts may reduce student autonomy or hinder higher-order thinking [42]. Our recent mapping study [43] found limited pedagogical grounding in prompt design for LLM-generated feedback. It also revealed diverse and evolving usage patterns, often without consistent theoretical foundations. Furthermore, institutions should establish clear ethical guidelines to prevent misuse and ensure that AI-generated feedback aligns with academic integrity standards [24], [27].

Despite these implications, this study has limitations. Being exploratory and discipline-specific (Computer Science), results might not generalize. Additionally, voluntary participation could introduce bias. The findings reflect usage patterns specific to the data collection period (August–December 2024), and newer AI models may shift user behaviors.

Future research should explore how advancements in generative AI (e.g., GPT-4.5, DeepSeek) influence user perceptions, adoption patterns, and educational outcomes. Longitudinal or comparative studies across different disciplines are recommended.

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